Excel & Power Query

Make everyday spreadsheet work faster.

Shortcut / reminder	Use it for
Ctrl + T	Turn a range into an Excel Table.
Ctrl + Shift + L	Toggle filters.
Ctrl + Arrow / Ctrl + Shift + Arrow	Jump to / select to the edge of a data region.
Ctrl + 1 / Ctrl + Alt + V	Format Cells / Paste Special.
F2 / F4	Edit a cell / cycle reference anchors while editing a formula.
Alt + = / Ctrl + `	AutoSum / show formulas.
Ctrl + ; / Ctrl + Alt + F5	Insert today's date / refresh all connections.
Alt, N, V	Open PivotTable options; press keys in sequence.

```
=SUMIFS(Sales[Revenue],Sales[Category],A2)
=COUNTIFS(Sales[Status],"Returned")
=XLOOKUP(A2,Products[ID],Products[Category],"Not found")
=IFERROR(B2/C2,"")
=TRIM(A2)
=MEDIAN(Sales[Revenue])
```

- Windows desktop shortcuts; laptop function keys may need Fn. Mac and web versions differ. XLOOKUP needs a supported Excel version.
- Power Query routine: connect → preserve raw data → set types and UK date locale → trim/clean → investigate blanks and duplicates → merge/append → reconcile → load → refresh.
- Merge adds columns by matching keys; append stacks rows. Check key uniqueness before a merge. Keep exception rows instead of silently deleting them.
- Formula examples assume the named tables and columns exist. IFERROR is not a substitute for checking the denominator.

References: Excel keyboard shortcuts · Excel PDF (CustomGuide)

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SQL & the T-SQL bridge

Remember the pattern, then check the result.

Shortcut / reminder	Use it for
SELECT / WHERE	Choose columns / filter rows before grouping.
GROUP BY / HAVING	Aggregate by a defined grain / filter groups.
LEFT JOIN	Keep every left-hand row; watch for one-to-many multiplication.
COUNT(*) / COUNT(column)	Count rows / count non-NULL values.
IS NULL / COALESCE	Test missing values / choose a deliberate replacement.
WITH name AS (...)	Name a CTE to make query stages readable.
ROW_NUMBER / RANK / DENSE_RANK	Unique sequence / ties with gaps / ties without gaps.
LAG / LEAD	Read a previous / next row within an ordered partition.

```
WITH monthly AS (
  SELECT category, SUM(revenue) AS sales
  FROM sales
  WHERE sale_date >= '2026-01-01'
    AND sale_date < '2026-02-01'
  GROUP BY category
)
SELECT category, sales,
  DENSE_RANK() OVER (ORDER BY sales DESC) AS sales_rank
FROM monthly;
```

- Logical order, simplified: FROM/JOIN → WHERE → GROUP BY → HAVING → window calculations → SELECT → DISTINCT → ORDER BY → row limit. Write a CTE when you need to filter a window result.
- T-SQL uses TOP (n) rather than PostgreSQL LIMIT n. Date functions, casts and identifier quoting vary; portable concepts do not make every query portable.
- Use an explicit ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW frame for a row-by-row running total; order by a stable tie-breaker.
- Before trusting a join: check key uniqueness, unmatched rows, row count and totals. Never use DISTINCT simply to conceal a broken join. Practise UPDATE/DELETE only in a safe copy.

References: SQL reference · T-SQL bridge · SQL essentials PDF (SQL Tutorial) · SQL window functions PDF (LearnSQL)

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Power BI, DAX & report operations

Fast authoring is useful. Reliable measures matter more.

Shortcut / reminder	Use it for
Shift + ?	Show available Power BI Desktop keyboard shortcuts.
Ctrl + F6 / Tab / Shift + Tab	Move between sections / forward / backward.
Alt	Show ribbon keytips.
Ctrl + Enter	Submit a DAX formula.
Ctrl + G	Go to a line in the DAX formula bar.
Alt + Up / Down	Move a line in the DAX formula bar.
Ctrl + G (canvas editing)	Group selected visuals; the same keys have a different formula-bar meaning.
Esc	Exit a visual or layer.

```
Revenue = SUM(Sales[Revenue])
Orders = DISTINCTCOUNT(Sales[OrderID])
Average Order Value = DIVIDE([Revenue], [Orders])
Revenue All Categories =
    CALCULATE([Revenue], REMOVEFILTERS(Products[Category]))
```

- Create these as measures; tables and columns are illustrative. A measure responds to filter context. A calculated column is evaluated row by row at refresh and stored.
- Model first: define fact-table grain, use dimension keys, prefer one-to-many relationships and deliberate filter direction. Confirm the date table before time-intelligence work.
- Release check: reconcile KPI totals → test slicers and blanks → add clear units/alt text → inspect mobile view → run Performance Analyzer → test RLS roles.
- Service checklist: workspace access, gateway where needed, credentials, scheduled refresh, refresh-failure ownership and sharing permissions. Desktop success is not proof of Service refresh or security.

References: Official Power BI keyboard shortcuts · Official DAX reference · Power BI & DAX PDF (Dataquest)

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Statistics, metrics & data quality

Keep the denominator, uncertainty and business question visible.

Shortcut / reminder	Use it for
Mean / median	Mean is sensitive to outliers; median is useful for skewed values.
Weighted mean	SUM(value × weight) / SUM(weight); validate the weights.
Percentage change	(new − old) / old; format as a percentage and define what happens at old = 0.
Percentage points	30% − 20% = 10 percentage points; the relative increase is 50%.
Conversion rate	Conversions / eligible opportunities over a defined period.
IQR	Q3 − Q1; inspect the distribution before using outlier rules.
Sample / population spread	Use STDEV.S for sample estimation; STDEV.P for the full population described.
Correlation	Association is not evidence of causation.

```
BEFORE ANALYSIS
Question → audience → metric → grain → period

AFTER EVERY TRANSFORMATION
Rows → keys → missingness → totals → exceptions

BEFORE SHARING
Source → assumptions → limits → privacy → owner
```

- A p-value is not the probability that the null hypothesis is true. Explain the test assumptions, effect size and uncertainty instead of treating 0.05 as an automatic business decision.
- A frequentist 95% confidence procedure covers the fixed true parameter in 95% of repeated samples under its assumptions; it is not a guarantee about an individual outcome.
- Missing is not automatically zero. Investigate causes and document any exclusion or imputation. Reconcile with an independent method where possible.
- Use only permitted public/synthetic practice data. Minimise personal information, keep raw data unchanged and document transformations and retention.

References: UK data-protection principles

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Python, pandas & plotting

Inspect first. Transform deliberately. Make the notebook reproducible.

Shortcut / reminder	Use it for
df.head() / df.info()	Preview rows / inspect types and missing counts.
df.isna().sum() / df.duplicated().sum()	Count missing values / exact duplicate rows.
df.loc[mask, columns]	Select explicit rows and columns; avoid chained assignment.
pd.to_datetime(..., errors="coerce")	Parse dates; inspect new NaT values instead of ignoring them.
groupby(...).agg(...)	Summarise at a defined grain.
merge(..., validate="many_to_one")	Enforce the expected join relationship.
pd.concat([...], ignore_index=True)	Append matching tables.
sort_values(...) / to_csv(..., index=False)	Order the result / export without the index.

```
import pandas as pd
import matplotlib.pyplot as plt

sales = pd.read_csv("sales.csv")
products = pd.read_csv("products.csv")
out = sales.merge(products, on="product_id",
    how="left", validate="many_to_one", indicator=True)
assert len(out) == len(sales)
print(out["_merge"].value_counts())
summary = out.groupby("category", dropna=False)["revenue"].sum()
summary.sort_values().plot.barh()
plt.xlabel("Revenue (£)")
plt.tight_layout()
plt.savefig("revenue_by_category.png", dpi=150)
```

- Example assumes the listed CSV columns exist and revenue is numeric. Resolve unmatched keys before interpreting category totals. Decide how missing revenue should be handled before summing.
- Create a virtual environment; install required libraries and record versions. Restart the kernel and run every cell in order before sharing.
- Python basics to recall: = assigns; == compares; indentation defines blocks; lists are ordered; dictionaries map keys to values. Use functions for repeated transformations.
- Name chart units, period and source. Use bars for category comparisons, lines for time series and scatter plots for relationships.

References: pandas API · Matplotlib cheat sheets · Python fundamentals PDF (Eric Matthes) · pandas PDF (pandas project)

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Setup, Git & README essentials

Keep the daily loop small and your public repository safe.

Shortcut / reminder	Use it for
git status / git diff	Inspect changed files / inspect unstaged changes.
git add file / git diff --staged	Stage a specific file / inspect what will be committed.
git commit -m "Message"	Save a descriptive local checkpoint.
git pull --rebase	Update from the tracked remote; only use when you understand local history and conflicts.
git push / git log --oneline	Publish commits / review compact history.
git restore --staged file	Unstage a file without discarding the working copy.
.gitignore	Exclude credentials, private data, environments and generated files.
Markdown: # / ## / - / [title](URL)	Title / subheading / list / descriptive link.

```
git status
git diff
git add analysis.sql README.md
git diff --staged
git commit -m "Add validated monthly sales analysis"
git pull --rebase
git push

# README headings
# Question | Data | Method | Checks
# Findings | Limits | Reproduce | Sources
```

- Set up folders for raw data, cleaned data, SQL, notebooks, reports and screenshots. Power BI Desktop needs Windows access. Check Excel/Power Query version support before starting.
- Do not use reset --hard, force push or history rewriting as a routine shortcut. Inspect files before staging; do not commit secrets or confidential data.
- If a secret is exposed, revoke/rotate it immediately; deleting the latest file does not remove earlier Git history.
- Add install/run steps, data source and licence, expected outputs and a screenshot. Check repository and demo links while logged out.

References: GitHub Hello World · Markdown reference · Git PDF (GitHub Education)

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Projects, capstone & handover

A repeatable workflow for every portfolio project.

Shortcut / reminder	Use it for
1 · Define	Write one useful question, its reader, the metric and success criteria.
2 · Source	Record source, licence, date, grain and limitations.
3 · Profile	Check types, ranges, blanks, duplicate keys and suspicious values.
4 · Transform	Keep raw inputs unchanged; document and reproduce each step.
5 · Validate	Reconcile counts, joins and key totals independently.
6 · Explain	Show three findings, an actionable recommendation and limits.
7 · Deliver	Package the workbook/query/report/notebook, README and screenshots.
8 · Handover	Explain refresh, access, definitions, exceptions and who owns failures.

```
CASE STUDY IN SIX LINES
Question: What decision does this support?
Data: Where did it come from and what does one row mean?
Method: What did I change or calculate?
Checks: How do I know the result is sound?
Finding: What changed, by how much, over which period?
Limit: What can this analysis not establish?
```

- Portfolio ladder: retail Excel → retail SQL → retail Power BI → public NHS pandas analysis → end-to-end regional capstone. Lead with your strongest three pieces.
- Public healthcare projects need dataset definitions, reporting-period rules and published methodology. They do not prove experience with internal clinical systems.
- Use screenshots, a PDF and a short demo if live sharing is unsuitable. Never use Publish to web for confidential or personal data.
- Label synthetic work honestly. Describe what you built and verified; do not claim fictional savings, workplace deployment or business impact.

References: Roadmap project workflow

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Revision, interviews & applications

Short preparation routines—not shortcuts around understanding.

Shortcut / reminder	Use it for
20-minute recall loop	5 minutes explain → 10 minutes solve → 5 minutes check and log one gap.
60-second introduction	Background → why analytics → evidence → role fit.
Project answer	Question → data → method → validation → finding → limitation.
Behavioural answer	Situation → task → action → result; describe your own contribution.
SQL test routine	Confirm grain and keys → write the query → test NULLs/ties → explain checks.
Role screening	Verify location, level, essential skills, deadline and genuine paid employment.
CV evidence	Tool + question answered + checks + truthful result; link the project.
Application tracking	Role, employer, URL, closing date, fit, tailored evidence, status and follow-up.

```
WEEKLY REVIEW
1. What can I now do without copying?
2. Which result have I independently checked?
3. What is the smallest remaining blocker?
4. Which role criteria can I demonstrate today?
5. What is my next repeatable practice session?
```

- Apply when you meet a role’s essential criteria; the roadmap week is not a ban on earlier applications. Excel-led reporting roles can have different requirements from SQL/BI roles.
- Keep one main resource per tool. After a lesson, close it and reproduce the task on a different small dataset. Extend the calendar when needed rather than abandoning the habit.
- Use truthful dates, responsibilities and results. A personal project is not employment experience and a certificate is not proof of independent delivery.
- During an interview, clarify assumptions, explain uncertainty and say how you would verify an unfamiliar detail. Do not bluff or use unauthorised assistance.

References: UK data analyst career guidance

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